EIN 4343 Inventory and Supply Chain Systems

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Newsvendor model: A critique

- No setup cost for placing an order
- Instantaneous order replenishment (i.e., zero order delivery lead time)

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A blueprint going forward

- Modeling of demand
  - Deterministic vs. stochastic

- Planning horizon length
  - Infinite vs. finite

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EIN4343 INVENTORY & SUPPLY CHAIN SYSTEMS

INVENTORY SYSTEMS: STOCHASTIC DEMAND INFINITE HORIZON MODEL

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(q, r) policy: Motivation

- Recall the EOQ model
  - Basic model: It was optimal to place an order for q units when the on-hand inventory dropped down to zero.
  - Positive delivery lead time extension: It was optimal to place an order for q units when inventory on-hand is sufficient to cover the demand during lead time (DDLT).

- Would this idea help us to handle situations when the demand is random?

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(q, r) policy: Assumptions

1. The inventory level is reviewed on a continuous basis.

2. The planning horizon is infinite.

3. The demand is random, and follows an underlying process with known parameters that do not change over time.

4. The entire order quantity can be delivered at the end of the lead time.

5. The order quantity is a continuous value.

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(q, r) policy: Assumptions

6. The system under consideration is a single item system.

7. The cost and revenue parameters are known and constant throughout the planning horizon.

8. There is a fixed cost associated with placing an order.

9. The order delivery lead-time is finite and known.

10. Shortages are allowed.

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(q, r) policy: Notation

c: unit procurement or purchase cost (\$/unit);
h: unit inventory holding cost (\$/unit/unit time);
b: unit shortage or penalty cost (\$/unit/unit time);
K: fixed order cost (\$/order);
X_L: random variable for demand during lead time (DDLT)(units);
$f(x_L)$: p. d. f. of X_L with expected value $E[X_L] = \mu$ and variance $Var(X_L) = \sigma^2$;

q: order quantity (units);
r: reorder level (units);
s: safety stock level (units);
T: inventory cycle length (time units); and

$C(q, r)$: the average expected total cost per unit time (\$/unit time).

NOTE: $f(x)$ is again distributed over the base $[0, \infty)$.

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Demand during lead time

- Suppose that the demand is a random variable denoted by X with mean and variance by μ_D and σ_D^2, respectively

- Suppose that the lead time is a constant parameter denoted by τ

Then, the demand during lead time (DDLT) is also a random variable denoted by X_L with mean and variance given by

$$\mu_L = \tau\mu_D$$
$$\sigma_L^2 = \tau\sigma_D^2$$

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Demand during lead time

- Suppose that the demand is a random variable denoted by X with mean and variance by μ_D and σ_D^2, respectively

- Suppose that the lead time is also a random variable denoted by T with mean and variance by μ_T and σ_T^2, respectively

Then, DDLT is again a random variable denoted by X_L with mean and variance given by

$$\mu_L = \mu_T \mu_D$$
$$\sigma_L^2 = \mu_T \sigma_D^2 + \mu_D^2 \sigma_T^2$$

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Example: Constant lead time

Consider a product for which the weekly demand is normally distributed with a mean of 500 units and a standard deviation of 50 units. If the lead time for the product is 6 weeks, what is the mean and standard deviation of DDLT?

$$\mu_L = \tau\mu_D = (6)(500) = 3{,}000 \text{ units}$$

$$\sigma_L^2 = \tau\sigma_D^2 = (6)(50)^2 = 15{,}000 \text{ units}^2$$

$$\Longrightarrow \sigma_L = 122.48 \text{ units}$$

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Example: Random lead time

Consider a product for which the weekly demand is normally distributed with a mean of 500 units and a standard deviation of 50 units. If the lead time for the product is a random variable with a mean of 6 weeks and a standard deviation of 2 weeks, what is the mean and standard deviation of DDLT?

$$\mu_L = \mu_D \mu_\tau = (6)(500) = 3{,}000 \text{ units}$$

$$\sigma_L^2 = \mu_T \sigma_D^2 + \mu_D^2 \sigma_T^2 = (6)(50)^2 + (500)^2 (2)^2 = 1{,}015{,}000 \text{ units}^2$$

$$\Longrightarrow \sigma_L = 1{,}007.47 \text{ units}$$

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Example: A comparison

KNOWN PARAMETER

When the lead time is constant, we have

$$\mu_L = 3{,}000 \text{ units}$$

$$\sigma_L = 122.48 \text{ units}$$

RANDOM VARIABLE

When the lead time is a random variable, we have

$$\mu_L = 3{,}000 \text{ units}$$

$$\sigma_L = 1{,}007.47 \text{ units}$$

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(q, r) policy: Excess demand

- Handling of excess demand has an impact on the analysis
  - Full backlogging
  - Lost sales

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EIN4343 INVENTORY & SUPPLY CHAIN SYSTEMS

INVENTORY SYSTEMS: STOCHASTIC DEMAND INFINITE HORIZON MODELS: FULL BACKLOGGING

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(q, r) policy: Full backlogging

- An important modeling primer…

- Expected safety stock level in this context
  - Not the layman's definition: Inventory we keep to hedge against uncertainty
  - But its analytical exposition: Amount of inventory on-hand when the next order is received

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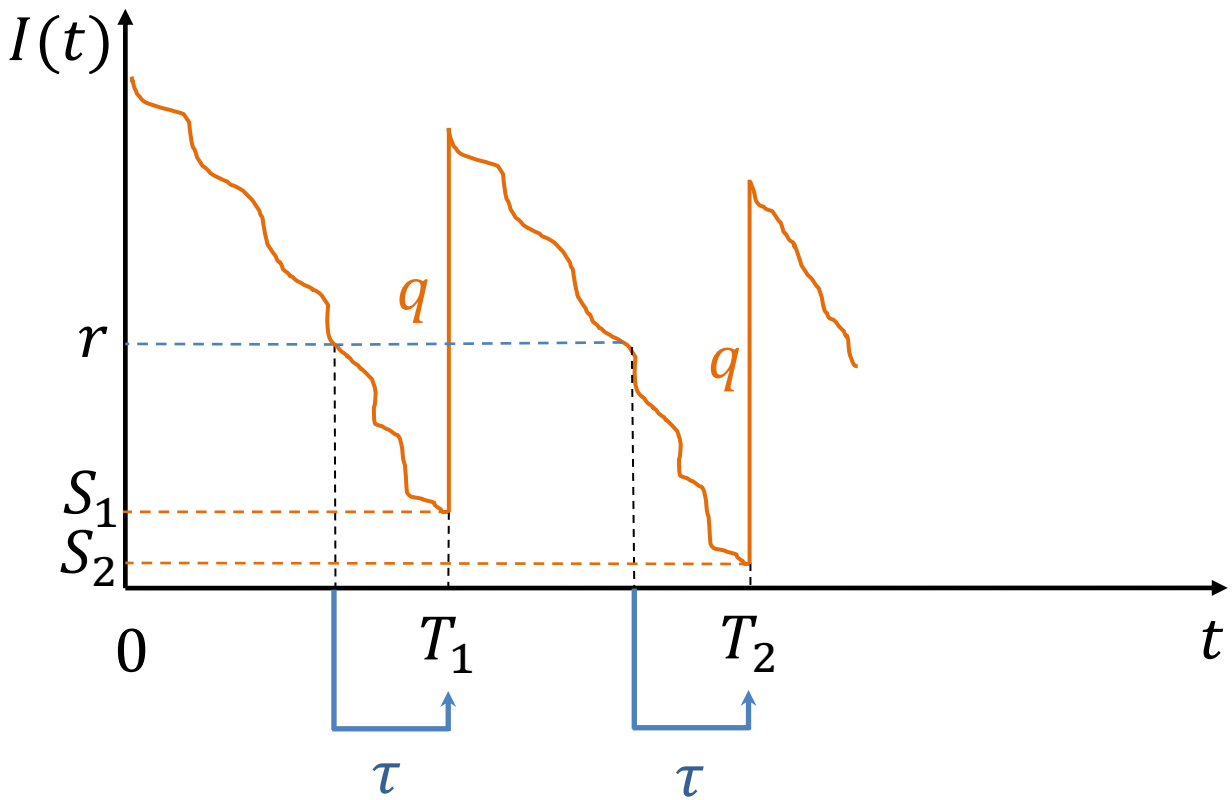
(q, r) policy: Full backlogging: Actual



Inventory cycle length T and the on-hand inventory level when an order arrives S (i.e., safety stock level) are random variables.

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A remark

Since it is infinite horizon problem, we will focus on the long-term average behavior of the system, i.e., expected values of all random variables namely

- demand rate,
- inventory cycle length, and
- safety stock level.

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(q, r) policy: Full backlogging: <u>Expected</u> demand rate

- Let λ denote the expected value of the annual demand (in units/year)

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(q, r) policy: Full backlogging: <u>Expected</u> inventory cycle length

- Let T denote the expected cycle length (in years)

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(q, r) policy: Full backlogging: <u>Expected</u> safety stock level

- Safety stock
  - Layman's definition
  - Technical definition
- Let s denote the expected on-hand inventory level when an order is delivered

$$s = \text{Reorder Level } - \text{Expected Demand During Lead Time}$$

$$s = r - E[X_L]$$

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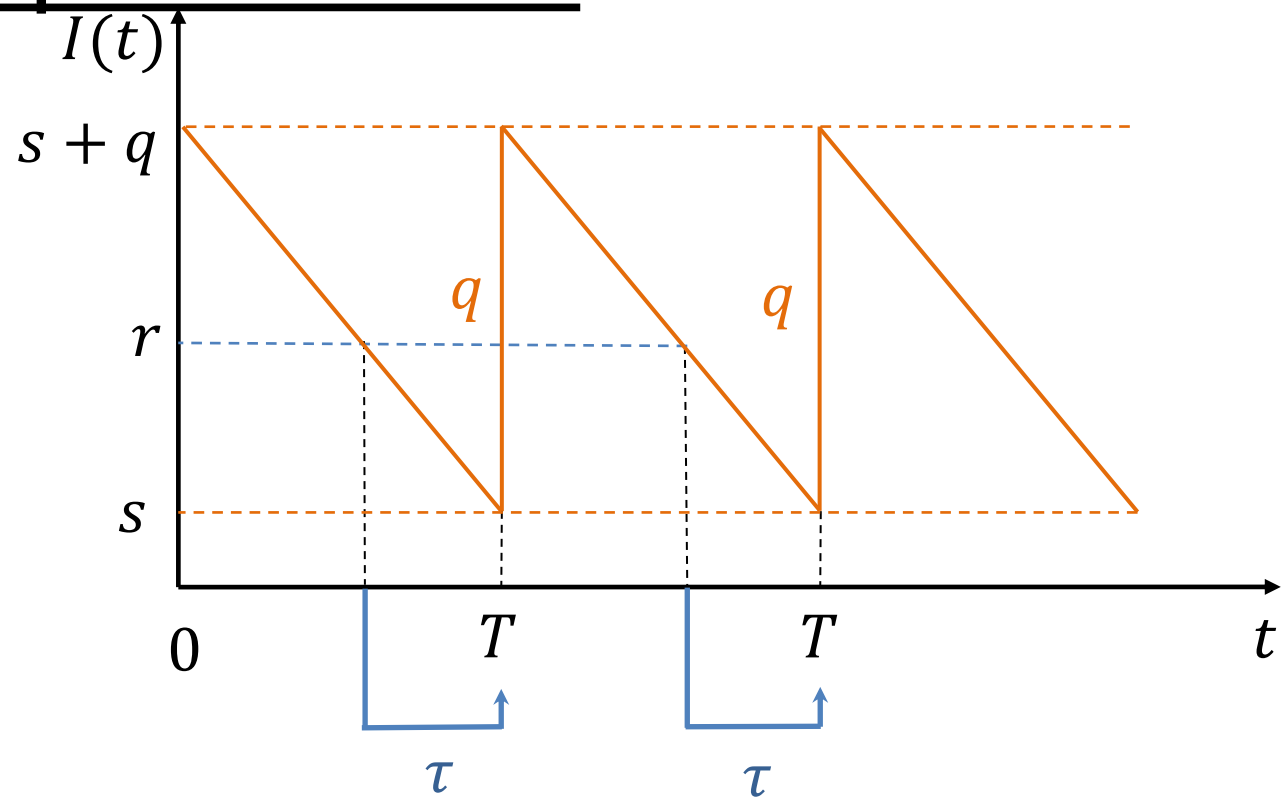
(q, r) policy: Full backlogging: Approximation



Inventory cycle length T and the on-hand inventory level when an order arrives s (i.e., safety stock level) are expected values of the corresponding random variables.

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(q, r) policy: Full backlogging: Approach

In order to minimize the average cost per unit time:

1. Determine the expected length of an inventory cycle

2. Calculate the expected total cost per inventory cycle

3. Divide the expected total cost per inventory cycle by the expected length of the inventory cycle to find the average expected total cost per unit time

4. Use continuous optimization techniques to determine the order quantity and the reorder level that jointly minimize the average expected total cost per unit time.

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(q, r) policy: Full backlogging: Expected cycle length

Expected cycle length is given by

$$T = \frac{q}{\lambda}$$

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(q, r) policy: Full backlogging: Expected total cost per cycle

Expected purchase cost per cycle

$$cq$$

Expected ordering cost per cycle

$$K$$

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(q, r) policy: Full backlogging: Expected total cost per cycle

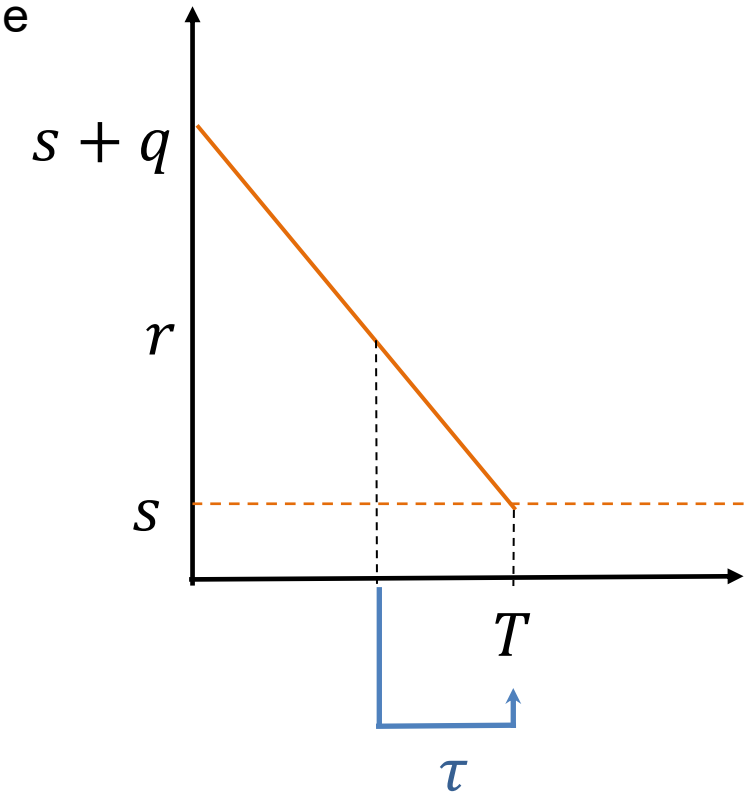
Expected inventory holding cost per cycle

$$h\int_0^{q/\lambda} I(t)dt = h\int_0^{q/\lambda}(q+s-\lambda t)dt$$

$$= h\left[(q+s)t - \frac{\lambda t^2}{2}\Big|_0^{q/\lambda}\right]$$

$$= h\left[\frac{(q+s)q}{\lambda} - \frac{q^2}{2\lambda}\right]$$

$$\equiv h\left[\frac{qT}{2} + sT\right]$$



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(q, r) policy: Full backlogging: Expected total cost per cycle

Expected backordering cost per cycle

$$bn(r)$$

where $n(r)$ represents the expected number of units that are backlogged per inventory cycle as a function of the reorder level r

$$n(r) = \int_r^{\infty} (x_L - r)\, f(x_L) dx_L$$

Note that this expression is valid for ANY probability distribution!

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(q, r) policy: Full backlogging: Expected total cost per cycle

Expected total cost per cycle is given by

$$cq + K + h\left[\frac{qT}{2} + sT\right] + bn(r)$$

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(q, r) policy: Full backlogging: Average expected total cost per unit time

$$C(q,r) = \frac{cq + K + h\left[\frac{qT}{2} + sT\right] + bn(r)}{T}$$

$$= c\lambda + \frac{K\lambda}{q} + h\left[\frac{q}{2} + (r - E[X_L])\right] + \frac{bn(r)\lambda}{q}$$

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(q, r) policy: Full backlogging: Optimization problem

$$\min C(q, r)$$

subject to

$$q, r \geq 0$$

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(q, r) policy: Full backlogging: First order conditions

$$C(q,r) = c\lambda + \frac{K\lambda}{q} + h\left[\frac{q}{2} + (r - E[X_L])\right] + \frac{bn(r)\lambda}{q}$$

$$\frac{\partial C}{\partial q} = -\frac{K\lambda}{q^2} + \frac{h}{2} - \frac{bn(r)\lambda}{q^2}$$

$$\frac{\partial C}{\partial r} = h + \frac{bn'(r)\lambda}{q}$$

$$n(r) = \int_r^{\infty} (x_L - r)\, f(x_L) dx_L \quad \Longrightarrow n'\ (r) = -(1 - F(r))$$

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(q, r) policy: Full backlogging: Optimal order quantity and reorder level

$$q^* = \sqrt{\frac{2\lambda[K+bn(r)]}{h}}$$

$$1 - F(r^*) = \frac{qh}{b\lambda}$$

Need to solve above two equations simultaneously to jointly optimize the order quantity and the reorder level.

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(q, r) policy: Full backlogging: Approach

Let j denote the iteration counter.

$$q^* = \sqrt{\frac{2\lambda[K+bn(r)]}{h}} \qquad (1)$$

$$1 - F(r^*) = \frac{qh}{b\lambda} \qquad (2)$$

1. Set $j = 0$. Initialize $q_j = \sqrt{\frac{2K\lambda}{h}}$
2. Evaluate r_j using q_j in (2). Go to step (5) if $r_j = r_{j-1}$
3. Use r_j to evaluate $n(r_j)$
4. Set $j = j + 1$. Evaluate q_j using $n(r_{j-1})$ in (1). Go to step (2) if $q_j \neq q_{j-1}$
5. Report $q^* = q_j$ and $r^* = r_j$

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(q, r) policy: Full backlogging: Evaluation of $n(r)$

If <u>DDLT is normally distributed</u>, $n(r)$ can be computed using the *standardized loss function* given by

$$L(z) = \int_z^{\infty} (t - z)\, \phi(t) dt$$

which represents the expected number of shortages as a fraction of the standard deviation of demand during lead time.

That is, we have

$$n(r) = \sigma_L L\left(\frac{r - \mu_L}{\sigma_L}\right)$$

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(q, r) policy: st

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Using standard normal and standard normal loss function tables

Recall the approach:

$$q^* = \sqrt{\frac{2\lambda[K+bn(r)]}{h}} \qquad (1)$$

$$1 - \boxed{F(r^*)} = \frac{qh}{b\lambda} \qquad (2)$$

Let j denote the iteration counter.

Use $F^{-1}(z_j)$ here to obtain z_j.

1. Set $j = 0$. Initialize $q_j = \sqrt{\frac{2K\lambda}{h}}$
2. Evaluate r_j using q_j in (2). Go to step (5) if $r_j = r_{j-1}$
3. Use r_j to evaluate $\boxed{n(r_j)}$ Use $L(z_j)$ here to obtain $n(r_j) = \sigma_L L\left(\frac{r_j - \mu_L}{\sigma_L}\right)$
4. Set $j = j + 1$. Evaluate q_j using $n(r_{j-1})$ in (1). Go to step (2) if $q_j \neq q_{j-1}$
5. Report $q^* = q_j$ and $\boxed{r^* = r_j}$ Use z^* here to obtain $r^* = \mu_L + z^*\sigma_L$.

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Identifying $F(z)$ and $L(z)$ values

Where are the

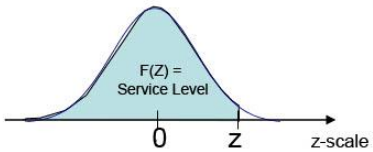
- z-values,
- $F(z)$ values, and
- $L(z)$ values?

Z-Chart & Loss Function

F(Z) is the probability that a variable from a standard normal distribution will be less than or equal to Z, or alternately, the service level for a quantity ordered with a z-value of Z.

L(Z) is the standard loss function, i.e. the expected number of lost sales as a fraction of the standard deviation. Hence, the lost sales = L(Z) x σ_{DEMAND}

Z	F(Z)	L(Z)	Z	F(Z)	L(Z)	Z	F(Z)	L(Z)	Z	F(Z)	L(Z)
-3.00	0.0013	3.000	-1.48	0.0694	1.511	0.04	0.5160	0.379	1.56	0.9406	0.026
-2.96	0.0015	2.960	-1.44	0.0749	1.474	0.08	0.5319	0.360	1.60	0.9452	0.023
-2.92	0.0018	2.921	-1.40	0.0808	1.437	0.12	0.5478	0.342	1.64	0.9495	0.021
-2.88	0.0020	2.881	-1.36	0.0869	1.400	0.16	0.5636	0.324	1.68	0.9535	0.019
-2.84	0.0023	2.841	-1.32	0.0934	1.364	0.20	0.5793	0.307	1.72	0.9573	0.017
-2.80	0.0026	2.801	-1.28	0.1003	1.327	0.24	0.5948	0.290	1.76	0.9608	0.016
-2.76	0.0029	2.761	-1.24	0.1075	1.292	0.28	0.6103	0.274	1.80	0.9641	0.014
-2.72	0.0033	2.721	-1.20	0.1151	1.256	0.32	0.6255	0.259	1.84	0.9671	0.013
-2.68	0.0037	2.681	-1.16	0.1230	1.221	0.36	0.6406	0.245	1.88	0.9699	0.012
-2.64	0.0041	2.641	-1.12	0.1314	1.186	0.40	0.6554	0.230	1.92	0.9726	0.010
-2.60	0.0047	2.601	-1.08	0.1401	1.151	0.44	0.6700	0.217	1.96	0.9750	0.009
-2.56	0.0052	2.562	-1.04	0.1492	1.117	0.48	0.6844	0.204	2.00	0.9772	0.008
-2.52	0.0059	2.522	-1.00	0.1587	1.083	0.52	0.6985	0.192	2.04	0.9793	0.008
-2.48	0.0066	2.482	-0.96	0.1685	1.050	0.56	0.7123	0.180	2.08	0.9812	0.007
-2.44	0.0073	2.442	-0.92	0.1788	1.017	0.60	0.7257	0.169	2.12	0.9830	0.006
-2.40	0.0082	2.403	-0.88	0.1894	0.984	0.64	0.7389	0.158	2.16	0.9846	0.005
-2.36	0.0091	2.363	-0.84	0.2005	0.952	0.68	0.7517	0.148	2.20	0.9861	0.005
-2.32	0.0102	2.323	-0.80	0.2119	0.920	0.72	0.7642	0.138	2.24	0.9875	0.004
-2.28	0.0113	2.284	-0.76	0.2236	0.889	0.76	0.7764	0.129	2.28	0.9887	0.004
-2.24	0.0125	2.244	-0.72	0.2358	0.858	0.80	0.7881	0.120	2.32	0.9898	0.003
-2.20	0.0139	2.205	-0.68	0.2483	0.828	0.84	0.7995	0.112	2.36	0.9909	0.003
-2.16	0.0154	2.165	-0.64	0.2611	0.798	0.88	0.8106	0.104	2.40	0.9918	0.003
-2.12	0.0170	2.126	-0.60	0.2743	0.769	0.92	0.8212	0.097	2.44	0.9927	0.002
-2.08	0.0188	2.087	-0.56	0.2877	0.740	0.96	0.8315	0.090	2.48	0.9934	0.002
-2.04	0.0207	2.048	-0.52	0.3015	0.712	1.00	0.8413	0.083	2.52	0.9941	0.002
-2.00	0.0228	2.008	-0.48	0.3156	0.684	1.04	0.8508	0.077	2.56	0.9948	0.002
-1.96	0.0250	1.969	-0.44	0.3300	0.657	1.08	0.8599	0.071	2.60	0.9953	0.001
-1.92	0.0274	1.930	-0.40	0.3446	0.630	1.12	0.8686	0.066	2.64	0.9959	0.001
-1.88	0.0301	1.892	-0.36	0.3594	0.605	1.16	0.8770	0.061	2.68	0.9963	0.001
-1.84	0.0329	1.853	-0.32	0.3745	0.579	1.20	0.8849	0.056	2.72	0.9967	0.001
-1.80	0.0359	1.814	-0.28	0.3897	0.554	1.24	0.8925	0.052	2.76	0.9971	0.001
-1.76	0.0392	1.776	-0.24	0.4052	0.530	1.28	0.8997	0.047	2.80	0.9974	0.001
-1.72	0.0427	1.737	-0.20	0.4207	0.507	1.32	0.9066	0.044	2.84	0.9977	0.001
-1.68	0.0465	1.699	-0.16	0.4364	0.484	1.36	0.9131	0.040	2.88	0.9980	0.001
-1.64	0.0505	1.661	-0.12	0.4522	0.462	1.40	0.9192	0.037	2.92	0.9982	0.001
-1.60	0.0548	1.623	-0.08	0.4681	0.440	1.44	0.9251	0.034	2.96	0.9985	0.000
-1.56	0.0594	1.586	-0.04	0.4840	0.419	1.48	0.9306	0.031	3.00	0.9987	0.000
-1.52	0.0643	1.548	0.00	0.5000	0.399	1.52	0.9357	0.028			



***Z* & *L(z)* for special service levels**

Service Level F(z)	*z*	*L(z)*
75%	0.67	0.150
90%	1.28	0.047
95%	1.64	0.021
99%	2.33	0.003

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(q, r) policy: Full backlogging: Example

Lowe's uses a continuous review system to track the sales and inventory levels of paint. Suppose that the weekly demand for a particular type of paint sold is approximately normally distributed with a mean of 40 cans and a standard deviation of 3 cans.

Suppose that the paint has a constant delivery lead time of 12 weeks, and the unit purchase cost is $7.20/can. Fixed cost of ordering is $750/order.

Unsatisfied demand can be backlogged into the next cycle. Lowe's estimates that the unit loss-of-goodwill cost associated with backlogging is $3/unit.

If the annual inventory carrying rate is 20 percent and Lowe's operates 52 weeks/year, determine the optimal order quantity and reorder level for Lowe's.

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(q, r) policy: Full backlogging: Solution approach

1. Determine the parameters of demand during lead time.

2. Determine the annual demand rate

3. Determine the annual inventory holding cost.

4. Initialize the order quantity.

5. Use the iterative approach.

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(q, r) policy: Full backlogging: Solution

1. Parameters of demand during lead time

$\mu_D = 40$ cans
$\sigma_D^2 = 3^2$ cans^2
$\mu_\tau = 12$ weeks
$\sigma_\tau^2 = 0^2$ weeks^2

$$\mu_L = \mu_\tau \mu_D = (12)(40) = 480 \text{ units}$$

$$\sigma_L^2 = \mu_\tau \sigma_D^2 + \mu_D^2 \sigma_\tau^2 = (12)(3^2) + (40^2)\,(0^2) \qquad = 108 \text{ units}^2$$

$$\Longrightarrow \sigma_L = 10.39 \text{ units}$$

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(q, r) policy: Full backlogging: Solution

$\mu_D = 40$ cans/week
52 weeks/year
$i = 0.20$
$c = \$7.20/\text{can}$

2. Annual demand rate

$$\lambda = (52)(40) = 2{,}080 \text{ units/year}$$

3. Unit inventory holding cost

$$h = ic = (0.2)(7.20) = \$1.44/\text{unit/year}$$

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(q, r) policy: Full backlogging: Solution: Iteration 0

$$q^* = \sqrt{\frac{2\lambda[K+bn(r)]}{h}} \qquad (1)$$

$$1 - F(r^*) = \frac{qh}{b\lambda} \qquad (2)$$

$$j = 0$$

$$q_0 = \sqrt{\frac{2\lambda K}{h}} = \sqrt{\frac{2(2{,}080)(750)}{1.44}} = 1{,}471.96 \text{ units}$$

Report two digits for z-value and three digits for L(z) value!

$$1 - F(r_0) = \frac{q_0 h}{b\lambda} \quad \Rightarrow F(r_0) = 1 - \frac{q_0 h}{b\lambda} = 1 - \frac{(1{,}471.96)(1.44)}{(3)(2{,}080)} = 0.6603$$

$$F(r_0) = 0.6603 \Rightarrow z_0 = 0.4133 \cong 0.41 \Rightarrow L(z_0) = 0.227$$

$$\frac{0.41 - 0.40}{0.44 - 0.40} = \frac{L(z_0) - 0.230}{0.217 - 0.230}$$

$$n(r_0) = \sigma_L L(z_0) = (10.39)(0.227) = 2.3585 \text{ units}$$

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(q, r) policy: Full backlogging: Solution: Iteration 1

$$q^* = \sqrt{\frac{2\lambda[K+bn(r)]}{h}} \qquad (1)$$

$$1 - F(r^*) = \frac{qh}{b\lambda} \qquad (2)$$

$j = 1$

$$q_1 = \sqrt{\frac{2\lambda[K + bn(r_0)]}{h}} = \sqrt{\frac{2(2{,}080)[750 + (3)(2.3585)]}{1.44}} = 1{,}478.89 \text{ units}$$

$$1 - F(r_1) = \frac{q_1 h}{b\lambda} \Rightarrow F(r_1) = 1 - \frac{q_1 h}{b\lambda} = 1 - \frac{(1{,}478.89)(1.44)}{(3)(2{,}080)} = 0.6588$$

$$F(r_1) = 0.6587 \Rightarrow z_1 = 0.4089 \cong 0.41 \Rightarrow L(z_1) = 0.227$$

$$\frac{0.41 - 0.40}{0.44 - 0.40} = \frac{L(z_1) - 0.230}{0.217 - 0.230}$$

$$n(r_1) = \sigma_L L(z_1) = (10.39)(0.227) = 2.3585 \text{ units}$$

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(q, r) policy: Full backlogging: Solution: Optimal solution

$n(r_1) = n(r_2)$ Convergence is achieved!

$$q^* = q_1 = 1{,}478.89 \text{ units}$$

$$z^* = z_1 = 0.41$$

$$r^* = r_1 = \mu_L + z^* \sigma_L = 480 + (0.41)(10.39) = 484.26 \text{ units}$$

Policy: Order 1,478.89 units when the on-hand inventory level drops down to 484.26 units

$$s^* = r^* - \mu_L = 484.26 - 480 = 4.26 \text{ units}$$

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Using standard normal and standard normal loss function tables

Recall the approach:

$$q^* = \sqrt{\frac{2\lambda[K+bn(r)]}{h}} \qquad (1)$$

$$1 - \boxed{F(r^*)} = \frac{qh}{b\lambda} \qquad (2)$$

Need $F^{-1}(z)$ here to obtain z.

Let j denote the iteration counter.

1. Set $j = 0$. Initialize $q_j = \sqrt{\frac{2K\lambda}{h}}$
2. Evaluate r_j using q_j in (2). Go to step (5) if $r_j = r_{j-1}$
3. Use r_j to evaluate $\boxed{n(r_j)}$ Use $L(z)$ here.
4. Set $j = j + 1$. Evaluate q_j using $n(r_{j-1})$ in (1). Go to step (2) if $q_j \neq q_{j-1}$
5. Report $q^* = q_j$ and $\boxed{r^* = r_j}$ Use z here.

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(q, r) policy: Lost sales

- Handling of excess demand has an impact on the analysis
  - Full backlogging
  - Lost sales

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EIN4343 INVENTORY & SUPPLY CHAIN SYSTEMS

INVENTORY SYSTEMS: STOCHASTIC DEMAND INFINITE HORIZON MODELS: LOST SALES

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A remark

Since it is infinite horizon problem, we will focus on the long-term average behavior of the system, i.e., expected values of all random variables namely

- demand rate, Same as before!
- inventory cycle length, and Same as before!
- safety stock level. Need to think about this more critically!

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(q, r) policy: Lost sales: <u>Expected</u> safety stock level

Let s denote the expected on-hand inventory level when an order is delivered

$$s = \int_0^r (r - x_L) f(x_L)\, dx_L$$

$$= \int_0^\infty (r - x_L) f(x_L)\, dx_L - \int_r^\infty (r - x_L) f(x_L)\, dx_L$$

$$= r \int_0^\infty f(x_L)\, dx_L - \int_0^\infty x_L f(x_L)\, dx_L - \int_r^\infty (r - x_L) f(x_L)\, dx_L$$

$$= r - E[X_L] - (-n(r))$$

$$= r - E[X_L] + n(r)$$

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(q, r) policy: Lost sales: Approach

In order to minimize the average cost per unit time:

1. Determine the expected length of an inventory cycle
2. Calculate the expected total cost per inventory cycle
3. Divide the expected total cost per inventory cycle by the expected length of the inventory cycle to find the average expected total cost per unit time
4. Use continuous optimization techniques to determine the order quantity and the reorder level that jointly minimize the average expected total cost per unit time.

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(q, r) policy: Lost sales: Expected cycle length

Expected cycle length is given by

$$T = \frac{q}{\lambda}$$

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(q, r) policy: Lost sales : Expected total cost per cycle

Expected purchase cost per cycle

$$cq$$

Expected ordering cost per cycle

$$K$$

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(q, r) policy: Lost sales : Expected total cost per cycle

Expected inventory holding cost per cycle

$$h\int_0^{q/\lambda} I(t)dt = h\left[\frac{qT}{2} + sT\right]$$

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(q, r) policy: Lost sales : Expected total cost per cycle

Expected shortage cost per cycle

$$bn(r)$$

where $n(r)$ represents the expected number of shortages per inventory cycle as a function of the reorder level r

$$n(r) = \int_r^{\infty} (x_L - r)\, f(x_L) dx_L$$

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(q, r) policy: Lost sales : Expected total cost per cycle

Expected total cost per cycle is given by

$$cq + K + h\left[\frac{qT}{2} + sT\right] + bn(r)$$

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(q, r) policy: Lost sales : Average expected total cost per unit time

$$C(q,r) = \frac{cq + K + h\left[\frac{qT}{2} + sT\right] + bn(r)}{T}$$

$$= c\lambda + \frac{K\lambda}{q} + h\left[\frac{q}{2} + (r - E[X_L] + n(r))\right] + \frac{bn(r)\lambda}{q}$$

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(q, r) policy: Lost sales : Optimization problem

$$\min C(q, r)$$

subject to

$$q, r \geq 0$$

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(q, r) policy: Lost sales : First order conditions

$$\mathrm{C}(q,r) = c\lambda + \frac{K\lambda}{q} + h\left[\frac{q}{2} + (r - E[X_L] + n(r))\right] + \frac{bn(r)\lambda}{q}$$

$$\frac{\partial C}{\partial q} = -\frac{K\lambda}{q^2} + \frac{h}{2} - \frac{bn(r)\lambda}{q^2}$$

$$\frac{\partial C}{\partial r} = h + hn'(r) + \frac{bn'(r)\lambda}{q}$$

$$n(r) = \int_r^\infty (x_L - r)\, f(x_L) dx_L \;\Longrightarrow n'\,(r) = -(1 - F(r))$$

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(q, r) policy: Lost sales : Optimal order quantity and reorder level

$$q^* = \sqrt{\frac{2\lambda[K+bn(r)]}{h}}$$

$$1 - F(r^*) = \frac{qh}{b\lambda+qh}$$

Need to solve above two equations simultaneously to jointly optimize the order quantity and the reorder level.

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(q, r) policy: Lost sales: Approach

Let j denote the iteration counter.

1. Set $j = 0$. Initialize $q_j = \sqrt{\frac{2K\lambda}{h}}$
2. Evaluate r_j using q_j in (4). Go to step (5) if $r_j = r_{j-1}$
3. Use r_j to evaluate $n(r_j)$
4. Set $j = j + 1$. Evaluate q_j using $n(r_{j-1})$ in (3). Go to step (2) if $q_j \neq q_{j-1}$
5. Report $q^* = q_j$ and $r^* = r_j$

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(q, r) policy: Lost sales: Example

Lowe's uses a continuous review system to track the sales and inventory levels of paint. Suppose that the weekly demand for a particular type of paint sold is approximately normally distributed with a mean of 40 cans and a standard deviation of 3 cans.

Suppose that the paint has a constant delivery lead time of 12 weeks, and the unit purchase cost is \$7.20/can. Fixed cost of ordering is \$750/order.

Unsatisfied demand can be backlogged into the next cycle. Lowe's estimates that the unit loss-of-goodwill cost associated with backlogging is \$3/unit.

Note that b is the same for demonstrative purposes!

If the annual inventory carrying rate is 20 percent and Lowe's operates 52 weeks/year, determine the optimal order quantity and reorder level for Lowe's.

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(q, r) policy: Lost sales: Solution: Iteration 0

$$q^* = \sqrt{\frac{2\lambda[K+bn(r)]}{h}} \qquad (3)$$

$$1 - F(r^*) = \frac{qh}{b\lambda+qh} \qquad (4)$$

$$j = 0$$

$$q_0 = \sqrt{\frac{2\lambda K}{h}} = \sqrt{\frac{2(52)(40)(750)}{(0.2)(7.2)}} = 1{,}471.96 \text{ units}$$

Report two digits for z-value and three digits for L(z) value!

$$1 - F(r_0) = \frac{q_0 h}{b\lambda + q_0 h}$$

$$\Rightarrow F(r_0) = 1 - \frac{q_0 h}{b\lambda + q_0 h} = 1 - \frac{(1{,}471.96)(1.44)}{(3)(2{,}080) + (1{,}471.96)(1.44)} = 0.7464$$

$$F(r_0) = 0.7464 \Rightarrow z_0 = 0.6632 \cong 0.66 \Rightarrow \mathrm{L}(z_0 = 0.66) = 0.153$$

$$n(r_0) = \sigma_L L(z_0) = (10.39)(0.153) = 1.59$$

$$\frac{0.66 - 0.64}{0.68 - 0.64} = \frac{L(z_0) - 0.158}{0.148 - 0.158}$$

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(q, r) policy: Lost sales: Solution: Iteration 1

$$q^* = \sqrt{\frac{2\lambda[K+bn(r)]}{h}} \qquad (3)$$

$$1 - F(r^*) = \frac{qh}{b\lambda+qh} \qquad (4)$$

$j = 1$

$$q_1 = \sqrt{\frac{2\lambda[K + bn(r_0)]}{h}} = \sqrt{\frac{2(2080)[750 + (3)(1.59)]}{1.44}} = 1{,}476.63 \text{ units}$$

$$1 - F(r_1) = \frac{q_1 h}{b\lambda + q_1 h} \Rightarrow F(r_1) = 1 - \frac{q_1 h}{b\lambda + q_1 h} = 0.7464$$

Basically, at this point we know that the solution has converged…

$$F(r_1) = 0.7464 \Rightarrow z_1 = 0.6632 \cong 0.66 \Rightarrow \mathrm{L}(0.66) = 0.153$$

$$n(r_1) = \sigma_L L(z_1) = (10.39)(0.153) = 1.59$$

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(q, r) policy: Lost sales: Solution: Optimal solution

$n(r_1) = n(r_2)$ Convergence is achieved!

$$q^* = q_1 = 1{,}476.63 \text{ units}$$

$$z^* = z_1 = 0.66$$

$$r^* = r_1 = \mu_L + z^* \sigma_L = 480 + (0.66)(10.39) = 486.86 \text{ units}$$

Policy: order 1,476.63 units when the on-hand inventory level drops down to 486.86 units

$$s^* = r^* - \mu_L + n(r^*) = 486.86 - 480 + 1.59 = 8.45 \text{ units}$$

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(q, r) policies : A comparison

Full backlogging

$q^* = 1{,}478.89$ units

$z^* = 0.41$

$r^* = 484.26$ units

Lost sales

$q^* = 1{,}476.63$ units

$z^* = 0.66$

$r^* = 486.86$ units

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Service levels in continuous review systems

- It is difficult to accurately estimate b, the unit cost per stock out.
  - Customer dissatisfaction
  - Impact on future sales
  - Inconvenience
  - Etc.

- It might be easier to take a product availability perspective.

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Measures of product availability

- *Cycle service level* (CSL) or *Type I service level* is the fraction of cycles for which all the demand is met. Or, the probability of not having a stock-out in an inventory cycle.

- *Product fill rate* (FR) or *Type II service level* is the fraction of product demand satisfied using product in inventory. Or, probability that demand is supplied from available inventory.

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Type I Service Level

- Control probability of not stocking out during lead time
  - Let α denote this probability

- Specification of α is sufficient to determine r and q can be determined independently

- Optimal policy under Type I

$$F(r) = \alpha$$

$$q = \sqrt{\frac{2K\lambda}{h}}$$

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Type II Service Level

- Control proportion of demand satisfied (immediately/directly) from stock
  - Let β denote this proportion

- Two approaches
  - Optimal policy – iterative approach
  - Approximate policy – one-step myopic approach

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Type II Service Level

- Optimal policy

$$n(r) = (1-\beta)q$$

$$q = \frac{n(r)}{1-F(r)} + \sqrt{\frac{2K\lambda}{h} + \left(\frac{n(r)}{1+F(r)}\right)^2}$$

- Approximate policy

$$n(r) = (1-\beta)q$$

$$q = \sqrt{\frac{2K\lambda}{h}}$$

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Service levels: Example

Lowe's uses a continuous review system to track the sales and inventory levels of paint. Suppose that the weekly demand for a particular type of paint sold is approximately normally distributed with a mean of 40 cans and a standard deviation of 3 cans.

Suppose that the paint has a constant delivery lead time of 12 weeks, and the unit purchase cost is $7.20/can. Fixed cost of ordering is $750/order.

The manager believes that the loss of goodwill cost cannot be estimated accurately and wants to use a service level of 99%.

If the annual inventory carrying rate is 20 percent and Lowe's operates 52 weeks/year, determine the policy parameters for Lowe's under (1) Type I service level and (2) Type II service level (you may use the approximate approach).

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Service levels: Solution Type I

$$q_{\alpha=0.99} = \sqrt{\frac{2K\lambda}{h}} = \sqrt{\frac{2(2{,}080)(750)}{1.44}} = 1{,}471.96 \text{ units}$$

$$F(r) = \alpha = 0.99 \Rightarrow z_{\alpha=0.99} = 2.33$$

$$r_{\alpha=0.99} = \mu_L + z_{\alpha=0.99}\sigma_L$$

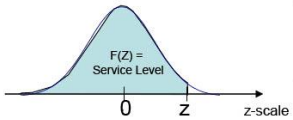
$$= 480 + (2.33)(10.39) = 504.21 \text{ units}$$

Z-Chart & Loss Function

F(Z) is the probability that a variable from a standard normal distribution will be less than or equal to Z, or alternately, the service level for a quantity ordered with a z-value of Z.

L(Z) is the standard loss function, i.e. the expected number of lost sales as a fraction of the standard deviation. Hence, the lost sales = L(Z) x σ_{DEMAND}

Z	F(Z)	L(Z)	Z	F(Z)	L(Z)	Z	F(Z)	L(Z)	Z	F(Z)	L(Z)
-3.00	0.0013	3.000	-1.48	0.0694	1.511	0.04	0.5160	0.379	1.56	0.9406	0.026
-2.96	0.0015	2.960	-1.44	0.0749	1.474	0.08	0.5319	0.360	1.60	0.9452	0.023
-2.92	0.0018	2.921	-1.40	0.0808	1.437	0.12	0.5478	0.342	1.64	0.9495	0.021
-2.88	0.0020	2.881	-1.36	0.0869	1.400	0.16	0.5636	0.324	1.68	0.9535	0.019
-2.84	0.0023	2.841	-1.32	0.0934	1.364	0.20	0.5793	0.307	1.72	0.9573	0.017
-2.80	0.0026	2.801	-1.28	0.1003	1.327	0.24	0.5948	0.290	1.76	0.9608	0.016
-2.76	0.0029	2.761	-1.24	0.1075	1.292	0.28	0.6103	0.274	1.80	0.9641	0.014
-2.72	0.0033	2.721	-1.20	0.1151	1.256	0.32	0.6255	0.259	1.84	0.9671	0.013
-2.68	0.0037	2.681	-1.16	0.1230	1.221	0.36	0.6406	0.245	1.88	0.9699	0.012
-2.64	0.0041	2.641	-1.12	0.1314	1.186	0.40	0.6554	0.230	1.92	0.9726	0.010
-2.60	0.0047	2.601	-1.08	0.1401	1.151	0.44	0.6700	0.217	1.96	0.9750	0.009
-2.56	0.0052	2.562	-1.04	0.1492	1.117	0.48	0.6844	0.204	2.00	0.9772	0.008
-2.52	0.0059	2.522	-1.00	0.1587	1.083	0.52	0.6985	0.192	2.04	0.9793	0.008
-2.48	0.0066	2.482	-0.96	0.1685	1.050	0.56	0.7123	0.180	2.08	0.9812	0.007
-2.44	0.0073	2.442	-0.92	0.1788	1.017	0.60	0.7257	0.169	2.12	0.9830	0.006
-2.40	0.0082	2.403	-0.88	0.1894	0.984	0.64	0.7389	0.158	2.16	0.9846	0.005
-2.36	0.0091	2.363	-0.84	0.2005	0.952	0.68	0.7517	0.148	2.20	0.9861	0.005
-2.32	0.0102	2.323	-0.80	0.2119	0.920	0.72	0.7642	0.138	2.24	0.9875	0.004
-2.28	0.0113	2.284	-0.76	0.2236	0.889	0.76	0.7764	0.129	2.28	0.9887	0.004
-2.24	0.0125	2.244	-0.72	0.2358	0.858	0.80	0.7881	0.120	2.32	0.9898	0.003
-2.20	0.0139	2.205	-0.68	0.2483	0.828	0.84	0.7995	0.112	2.36	0.9909	0.003
-2.16	0.0154	2.165	-0.64	0.2611	0.798	0.88	0.8106	0.104	2.40	0.9918	0.003
-2.12	0.0170	2.126	-0.60	0.2743	0.769	0.92	0.8212	0.097	2.44	0.9927	0.002
-2.08	0.0188	2.087	-0.56	0.2877	0.740	0.96	0.8315	0.090	2.48	0.9934	0.002
-2.04	0.0207	2.048	-0.52	0.3015	0.712	1.00	0.8413	0.083	2.52	0.9941	0.002
-2.00	0.0228	2.008	-0.48	0.3156	0.684	1.04	0.8508	0.077	2.56	0.9948	0.002
-1.96	0.0250	1.969	-0.44	0.3300	0.657	1.08	0.8599	0.071	2.60	0.9953	0.001
-1.92	0.0274	1.930	-0.40	0.3446	0.630	1.12	0.8686	0.066	2.64	0.9959	0.001
-1.88	0.0301	1.892	-0.36	0.3594	0.605	1.16	0.8770	0.061	2.68	0.9963	0.001
-1.84	0.0329	1.853	-0.32	0.3745	0.579	1.20	0.8849	0.056	2.72	0.9967	0.001
-1.80	0.0359	1.814	-0.28	0.3897	0.554	1.24	0.8925	0.052	2.76	0.9971	0.001
-1.76	0.0392	1.776	-0.24	0.4052	0.530	1.28	0.8997	0.047	2.80	0.9974	0.001
-1.72	0.0427	1.737	-0.20	0.4207	0.507	1.32	0.9066	0.044	2.84	0.9977	0.001
-1.68	0.0465	1.699	-0.16	0.4364	0.484	1.36	0.9131	0.040	2.88	0.9980	0.001
-1.64	0.0505	1.661	-0.12	0.4522	0.462	1.40	0.9192	0.037	2.92	0.9982	0.001
-1.60	0.0548	1.623	-0.08	0.4681	0.440	1.44	0.9251	0.034	2.96	0.9985	0.000
-1.56	0.0594	1.586	-0.04	0.4840	0.419	1.48	0.9306	0.031	3.00	0.9987	0.000
-1.52	0.0643	1.548	0.00	0.5000	0.399	1.52	0.9357	0.028			



***Z* & *L(z)* for special service levels**

Service Level F(z)	*z*	*L(z)*
75%	0.67	0.150
90%	1.28	0.047
95%	1.64	0.021
99%	2.33	0.003

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Service levels: Solution Type II

$$q_{\beta=0.99} = \sqrt{\frac{2K\lambda}{h}} = \sqrt{\frac{2(2{,}080)(750)}{1.44}} = 1{,}471.96 \text{ units}$$

$$n(r) = (1-\beta)q = (1-0.99)(1{,}471.96) = 14.72 \text{ units}$$

$$n(r) = \sigma_L L\left(z_{\beta=0.99}\right) \Rightarrow L\left(z_{\beta=0.99}\right) = \frac{n(r)}{\sigma_L} = \frac{14.72}{10.39}$$

$$L\left(z_{\beta=0.99}\right) = 1.417 \Rightarrow z_{\beta=0.99} = -1.38$$

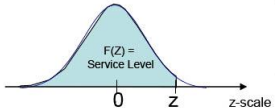
$$r_{\beta=0.99} = \mu_L + z_{\beta=0.99}\sigma_L$$
$$= 480 + (-1.38)(10.39) = 465.66 \text{ units}$$

Z-Chart & Loss Function

F(Z) is the probability that a variable from a standard normal distribution will be less than or equal to Z, or alternately, the service level for a quantity ordered with a z-value of Z.

L(Z) is the standard loss function, i.e. the expected number of lost sales as a fraction of the standard deviation. Hence, the lost sales = L(Z) x σ_{DEMAND}

Z	F(Z)	L(Z)	Z	F(Z)	L(Z)	Z	F(Z)	L(Z)	Z	F(Z)	L(Z)
-3.00	0.0013	3.000	-1.48	0.0694	1.511	0.04	0.5160	0.379	1.56	0.9406	0.026
-2.96	0.0015	2.960	-1.44	0.0749	1.474	0.08	0.5319	0.360	1.60	0.9452	0.023
-2.92	0.0018	2.921	-1.40	0.0808	1.437	0.12	0.5478	0.342	1.64	0.9495	0.021
-2.88	0.0020	2.881	-1.36	0.0869	1.400	0.16	0.5636	0.324	1.68	0.9535	0.019
-2.84	0.0023	2.841	-1.32	0.0934	1.364	0.20	0.5793	0.307	1.72	0.9573	0.017
-2.80	0.0026	2.801	-1.28	0.1003	1.327	0.24	0.5948	0.290	1.76	0.9608	0.016
-2.76	0.0029	2.761	-1.24	0.1075	1.292	0.28	0.6103	0.274	1.80	0.9641	0.014
-2.72	0.0033	2.721	-1.20	0.1151	1.256	0.32	0.6255	0.259	1.84	0.9671	0.013
-2.68	0.0037	2.681	-1.16	0.1230	1.221	0.36	0.6406	0.245	1.88	0.9699	0.012
-2.64	0.0041	2.641	-1.12	0.1314	1.186	0.40	0.6554	0.230	1.92	0.9726	0.010
-2.60	0.0047	2.601	-1.08	0.1401	1.151	0.44	0.6700	0.217	1.96	0.9750	0.009
-2.56	0.0052	2.562	-1.04	0.1492	1.117	0.48	0.6844	0.204	2.00	0.9772	0.008
-2.52	0.0059	2.522	-1.00	0.1587	1.083	0.52	0.6985	0.192	2.04	0.9793	0.008
-2.48	0.0066	2.482	-0.96	0.1685	1.050	0.56	0.7123	0.180	2.08	0.9812	0.007
-2.44	0.0073	2.442	-0.92	0.1788	1.017	0.60	0.7257	0.169	2.12	0.9830	0.006
-2.40	0.0082	2.403	-0.88	0.1894	0.984	0.64	0.7389	0.158	2.16	0.9846	0.005
-2.36	0.0091	2.363	-0.84	0.2005	0.952	0.68	0.7517	0.148	2.20	0.9861	0.005
-2.32	0.0102	2.323	-0.80	0.2119	0.920	0.72	0.7642	0.138	2.24	0.9875	0.004
-2.28	0.0113	2.284	-0.76	0.2236	0.889	0.76	0.7764	0.129	2.28	0.9887	0.004
-2.24	0.0125	2.244	-0.72	0.2358	0.858	0.80	0.7881	0.120	2.32	0.9898	0.003
-2.20	0.0139	2.205	-0.68	0.2483	0.828	0.84	0.7995	0.112	2.36	0.9909	0.003
-2.16	0.0154	2.165	-0.64	0.2611	0.798	0.88	0.8106	0.104	2.40	0.9918	0.003
-2.12	0.0170	2.126	-0.60	0.2743	0.769	0.92	0.8212	0.097	2.44	0.9927	0.002
-2.08	0.0188	2.087	-0.56	0.2877	0.740	0.96	0.8315	0.090	2.48	0.9934	0.002
-2.04	0.0207	2.048	-0.52	0.3015	0.712	1.00	0.8413	0.083	2.52	0.9941	0.002
-2.00	0.0228	2.008	-0.48	0.3156	0.684	1.04	0.8508	0.077	2.56	0.9948	0.002
-1.96	0.0250	1.969	-0.44	0.3300	0.657	1.08	0.8599	0.071	2.60	0.9953	0.001
-1.92	0.0274	1.930	-0.40	0.3446	0.630	1.12	0.8686	0.066	2.64	0.9959	0.001
-1.88	0.0301	1.892	-0.36	0.3594	0.605	1.16	0.8770	0.061	2.68	0.9963	0.001
-1.84	0.0329	1.853	-0.32	0.3745	0.579	1.20	0.8849	0.056	2.72	0.9967	0.001
-1.80	0.0359	1.814	-0.28	0.3897	0.554	1.24	0.8925	0.052	2.76	0.9971	0.001
-1.76	0.0392	1.776	-0.24	0.4052	0.530	1.28	0.8997	0.047	2.80	0.9974	0.001
-1.72	0.0427	1.737	-0.20	0.4207	0.507	1.32	0.9066	0.044	2.84	0.9977	0.001
-1.68	0.0465	1.699	-0.16	0.4364	0.484	1.36	0.9131	0.040	2.88	0.9980	0.001
-1.64	0.0505	1.661	-0.12	0.4522	0.462	1.40	0.9192	0.037	2.92	0.9982	0.001
-1.60	0.0548	1.623	-0.08	0.4681	0.440	1.44	0.9251	0.034	2.96	0.9985	0.000
-1.56	0.0594	1.586	-0.04	0.4840	0.419	1.48	0.9306	0.031	3.00	0.9987	0.000
-1.52	0.0643	1.548	0.00	0.5000	0.399	1.52	0.9357	0.028			



***Z* & *L(z)* for special service levels**

Service Level F(z)	z	L(z)
75%	0.67	0.150
90%	1.28	0.047
95%	1.64	0.021
99%	2.33	0.003

Is obtained by interpolation using

$$\frac{-1.40 - z}{-1.40 - (-1.36)} = \frac{1.437 - 1.417}{1.437 - 1.400}$$

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Service Levels: A comparison

Type I

$q_{\alpha=0.99} = 1{,}471.96 \text{ units}$

$r_{\alpha=0.99} = 504.21 \text{ units}$

Type II (approximate)

$q_{\beta=0.99} = 1{,}471.96 \text{ units}$

$r_{\beta=0.99} = 465.66 \text{ units}$

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What matters?

1. Computing the parameters of a (q, r) policy

2. Computing the expected safety inventory level (s) given the parameters of a (q, r) policy

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What matters?

3. Computing the parameters of a (q, r) policy and safety inventory (s) given a desired cycle service level (CSL, i.e., Type 1)

4. Computing the implied cycle service level (CSL, i.e., Type 1) given the parameters of a (q, r) policy

5. Computing parameters of a (q, r) policy and safety inventory (s) given a desired fill rate(FR , i.e., Type II)

6. Computing the implied fill rate (FR, i.e., Type II) given the parameters of a (q, r) policy

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Example 1: Statement

Consider an item for which the demand per week is approximately normally distributed with a mean of 80 units and a standard deviation of 40 units. This item is replenished from a foreign supplier and the constant lead time is 12 weeks.

Currently, the store manager orders 2,000 units when the on-hand inventory level drops down to 1,200 units.

What is the Type I Service Level achieved by the current order policy?

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Example 1: Analysis

What type of a problem is this?

What are the parameters of demand during lead time?

$\mu_L = (12)(80) \ = 960\ units$

$\sigma_L = \sqrt{12}(40) = 138.56\ units$

What are the parameters of the current inventory policy?

$q_{\alpha=x} = 2{,}000\ units$

$r_{\alpha=x} = 1{,}200\ unit$

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Example 1: Solution

$$F(r) = \alpha = P\{X_L \leq r\} = P\left\{\frac{x_L - \mu_L}{\sigma_L} \leq \frac{r - \mu_L}{\sigma_L}\right\} = P\{Z \leq z\} = \Phi(z)$$

$$z = \frac{r - \mu_L}{\sigma_L} = \frac{1{,}200 - 960}{138.56} = 1.7321$$

$$\frac{.9608 - \alpha}{9.608 - .9573} = \frac{1.76 - 1.7321}{1.76 - 1.72} \qquad \Longrightarrow \alpha = 0.9584$$

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Example 1: Additional insight

Order quantity (q)	Reorder level (r)	Z-value (z)	Implied service level (α)
2,000	1,350	2.8147	0.9976
2,000	1,300	2.4538	0.9929
2,000	1,250	2.0930	0.9818
2,000	1,200	1.7321	0.9584
2,000	1,150	1.3713	0.9149
2,000	1,100	1.0104	0.8438
2,000	1,050	0.6495	0.7420
2,000	1,000	0.2887	0.6136
2,000	950	-0.07217	0.4712

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Example 2: Statement

Consider an item for which the demand per week is approximately normally distributed with a mean of 80 units and a standard deviation of 40 units. This item is replenished from a foreign supplier and the constant lead time is 6 weeks.

Currently, the store manager orders 2,000 units when the on-hand inventory level drops down to 1,200 units.

What is the Type II Service Level achieved by the current order policy?

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Example 2: Analysis

What type of a problem is this?

What are the parameters of demand during lead time?

$\mu_L = (12)(80) = 960\ units$

$\sigma_L = \sqrt{12}(40) = 138.56\ units$

What are the parameters of the current inventory policy?

$q_{\beta=x} = 2{,}000\ units$

$r_{\beta=x} = 1{,}200\ units$

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Example 2: Solution

$$(1-\beta)q = n(r)$$

$$n(r) = \sigma_L L(z) \ = \sigma_L L\left(\frac{r-\mu_L}{\sigma_L}\right)$$

$$z = \frac{r-\mu_L}{\sigma_L} = \frac{1{,}200-960}{138.56} = 1.7321$$

$$\frac{1.7321-1.72}{1.76-1.72} = \frac{L(z)-0.017}{0.016-0.017} \quad \Longrightarrow L(z) = 0.0167$$

$$\Longrightarrow n(r) = \sigma_L L(z) = (138.56)(0.0167) = 2.3140\ units$$

$$\Longrightarrow \beta = 1 - \frac{n(r)}{q} = 1 - \frac{2.3140}{2000} = 0.9988$$

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Example 1: Additional insight

Order quantity (q)	Reorder level (r)	Z-value (z)	L(z) value	Implied service level (α)
2,000	1,200	1.7321	0.0167	0.9988
1,500	1,200	1.7321	0.0167	0.9985
1,000	1,200	1.7321	0.0167	0.9977

Order quantity (q)	Reorder level (r)	Z-value (z)	L(z) value	Implied service level (α)
2,000	1,200	1.7321	0.0167	0.9988
2,000	1,000	0.2887	0.2713	0.9812
2,000	800	-1.1547	1.0415	0.9278

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